

Abhinav Gyawali

✉ abhizer@abhizer.com | [github](https://github.com/abhizer) | [in](https://www.linkedin.com/in/abhizer) | [f](https://www.feldera.com/abhizer) | 📍 Austin, TX

Research Interests

Database systems, streaming and incremental computation, fault-tolerant distributed systems, and unstructured data analysis.

Education

University of Texas at Austin PhD in Computer Science <i>Advisors:</i> Dixin Tang, Vijay Chidambaram <i>Fellowship:</i> UTCS Distinguished Graduate Fellowship	Aug 2026 – present
Tribhuvan University BS in Computer Science and Information Technology	Oct 2019 – Aug 2024

Publications

Mihai Budiu, Leonid Ryzhyk, Gerd Zellweger, Ben Pfaff, Lalith Suresh, Simon Kassing, Abhinav Gyawali , Matei Budiu, Tej Chajed, Frank McSherry, Val Tannen. DBSP: Automatic Incremental View Maintenance for Rich Query Language. <i>The VLDB Journal</i> , 2025. (The “Best of VLDB 2023” special issue.)	VLDBJ 2025
--	------------

Research & Industry Experience

Feldera Software Engineer - Designed and implemented a distributed, fault-tolerant pipeline execution system with S3-backed checkpointing, crash recovery, and real-time output connectors for Postgres and Redis. - Extended Feldera’s SQL compiler with support for complex types (VARIANT); built automated differential testing infrastructure using SQLancer 🔗, uncovering 31 previously undetected bugs. - Developed the Feldera Python SDK 🔗 for programmatic pipeline control and authored technical documentation covering stream processing, batch processing, and observability.	US, Remote Dec 2023 – July 2026
Invisid Software Developer - Developed trackpad-rs 🔗, a Rust library for high-fidelity trackpad and mouse interaction data from Apple’s <i>MultitouchSupport</i> framework, enabling continuous biometric authentication. - Redesigned backend server infrastructure for behavioral authentication, improving efficiency, scalability, and consistency of biometric data processing.	Germany, Remote May 2022 – Nov 2023
Flow Webinar Security Engineer (Part-time) Conducted vulnerability assessment and penetration testing of a WebRTC-based video conferencing platform; managed backend infrastructure and system administration for production deployments.	US, Remote May 2020 – June 2021

Selected Projects

DBSP Rust Incremental computation engine where change propagation runs in time proportional to the change size rather than the full dataset – the engine underlying the VLDBJ publication.	github feldera/dbsp 🔗
---	--

SocketDB | Rust

- Lightweight SQL database that surfaces real-time query updates to clients over WebSockets.

 [abhizer/socketdb](#)

Nyx-lang | Rust

- Statically typed, tree-walking interpreted language with a full type-checking pass for safety and correctness.

 [abhizer/nyx-lang](#)

Loogle-rs | Rust

- Local full-text search engine based on TF-IDF ranking.

 [abhizer/loogle-rs](#)

Honors & Awards

UTCS Distinguished Graduate Fellowship | University of Texas at Austin

2026

SIGPLAN-M Mentee | Mentor: Prof. Jonathan Balkind

2025

DWIT Merit-Based Scholarship | Deerwalk Institute of Technology

2020, 2021